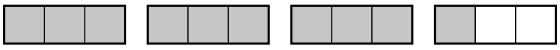
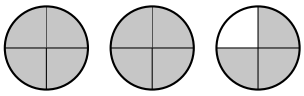
Lesson 1: Fractions Refresher: Mixed Numbers

Assignment

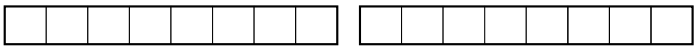
1. Write each picture as a mixed number and as an improper fraction.

(a) 

Mixed: _____ Improper: _____

(b) 

Mixed: _____ Improper: _____

(c) Shade the bars to show $1\frac{5}{8}$. 

2. Write each improper fraction as a mixed number (or a whole number) in simplest form.

(a) $\frac{9}{2}$ (b) $\frac{22}{5}$ (c) $\frac{31}{6}$ (d) $\frac{50}{8}$ (e) $\frac{27}{9}$ (f) $\frac{44}{12}$

3. Write each mixed number as an improper fraction.

(a) $2\frac{1}{4}$ (b) $3\frac{2}{5}$ (c) $5\frac{3}{8}$ (d) $7\frac{5}{6}$ (e) $10\frac{3}{4}$ (f) $8\frac{11}{16}$

4. A pizza place cuts every pizza into 8 slices.

(a) At a class party, students ate 29 slices. How many pizzas is that, as a mixed number?

(b) The hockey team ate $4\frac{3}{4}$ pizzas. How many slices is that?

5. Add. Write each answer in simplest form.

(a) $\frac{2}{5} + \frac{1}{5}$ (b) $\frac{1}{2} + \frac{3}{8}$ (c) $3\frac{1}{4} + 2\frac{1}{4}$

(d) $1\frac{2}{3} + 4\frac{1}{6}$ (e) $6\frac{3}{4} + 2\frac{5}{8}$ (f) $3\frac{4}{5} + 2\frac{1}{2}$

6. Subtract. Write each answer in simplest form.

(a) $5\frac{7}{8} - 1\frac{3}{8}$ (b) $4\frac{5}{6} - 2\frac{1}{3}$ (c) $6\frac{1}{4} - 3\frac{3}{4}$

(d) $8 - 2\frac{3}{5}$ (e) $7\frac{1}{3} - 4\frac{3}{4}$ (f) $10\frac{1}{8} - 5\frac{1}{2}$

7. A wall is framed with wood studs that are $3\frac{1}{2}$ in thick. It has $\frac{1}{2}$ in drywall on one side and $\frac{5}{8}$ in drywall on the other side.

(a) How thick is the finished wall?

(b) A door frame is $4\frac{9}{16}$ in deep. Is it deeper or shallower than the wall? By how much?

8. Lucas has a 72 in board. He cuts off one piece $29\frac{5}{8}$ in long and another piece $17\frac{3}{4}$ in long. Ignoring the saw cuts, how much of the board is left?

9. Multiply. Write each answer in simplest form.

(a) $\frac{3}{4} \times \frac{2}{3}$ (b) $3 \times 1\frac{2}{3}$ (c) $6 \times 2\frac{3}{4}$

(d) $\frac{1}{2} \times 5\frac{1}{4}$ (e) $\frac{2}{3}$ of $4\frac{1}{2}$ (f) $2\frac{1}{3} \times 1\frac{1}{2}$

10. Mia worked out $3\frac{1}{2} \times 2\frac{1}{4}$ like this: $3 \times 2 = 6$ and $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$, so the answer is $6\frac{1}{8}$.

(a) Use an estimate to show that her answer is too small.

(b) Find the correct answer.

11. A muffin recipe makes 12 muffins. Complete the table to make 18 muffins ($1\frac{1}{2}$ batches) and 6 muffins ($\frac{1}{2}$ batch).

Ingredient	**12 muffins**	**18 muffins**	**6 muffins**
Flour	$2\frac{1}{4}$ cups		
Brown sugar	$\frac{2}{3}$ cup		
Milk	$1\frac{1}{4}$ cups		
Butter	$\frac{1}{3}$ cup		

12. Grace is sewing 5 cushion covers. Each cover needs $1\frac{3}{8}$ yd of fabric. She bought 8 yd.

(a) How much fabric does she need for the 5 covers?

(b) How much fabric will be left over?

(c) Is the leftover fabric enough for a sixth cover? Explain.

13. A landscaper puts plastic edging along three garden beds. The beds need $12\frac{3}{4}$ ft, $9\frac{5}{8}$ ft and $15\frac{1}{2}$ ft of edging.

(a) Estimate the total length by rounding each one to the nearest foot.

(b) Find the exact total. Check it with your calculator.

(c) Edging is sold in 20 ft rolls. How many rolls does the landscaper need? Explain.

14. A shop drawing gives some lengths as decimals. Write each one as a mixed number of inches in simplest form, and write (d) as a decimal.

(a) 2.75 in (b) 5.125 in (c) 1.4375 in (d) $6\frac{3}{8}$ in

15. Ravi is building a fence section from 14 boards. Each board is $5\frac{1}{2}$ in wide. He leaves a $\frac{3}{4}$ in gap between each pair of boards (there is no gap at the ends). How long is the fence section?

16. What is $2\dfrac{3}{5}$ as an improper fraction?

A. $\dfrac{11}{5}$ **B.** $\dfrac{13}{5}$ **C.** $\dfrac{10}{5}$ **D.** $\dfrac{6}{5}$

17. $6\dfrac{3}{8} - 2\dfrac{5}{8} =$

A. $4\dfrac{1}{4}$ **B.** $3\dfrac{3}{8}$ **C.** $4\dfrac{3}{4}$ **D.** $3\dfrac{3}{4}$

18. Numerical Response Chris rides his bike $3\frac{3}{4}$ km to school and the same distance home, 5 days a week. How many kilometres does he ride in a week? Write your answer as a decimal.

Lesson 2: The SI System: Units, Prefixes and Referents

Assignment

1. For each item, write what is being measured (length, mass, capacity, temperature or area) and the SI unit you would use.

Item	What is measured?	SI unit (symbol)
(a) the length of a swimming pool		
(b) how much milk a jug holds		
(c) how heavy a backpack is		
(d) how cold it is inside a fridge		
(e) the lawn to be covered with sod		
(f) the height of a tree		

2.

(a) Complete the table.

Prefix	Symbol	Worth
kilo	k	1000
hecto		
deca		
deci		
centi		
milli		

(b) Write the full name of each unit.

km _____ mg _____

cL _____ dam _____

(c) Write the symbol for each unit.

millimetre _____ kilogram _____

millilitre _____ centimetre _____

3. Complete each statement.

(a) 1 m = _____ cm (b) 1 km = _____ m (c) 1 cm = _____ mm (d) 1 m = _____ mm

(e) 1 kg = _____ g (f) 1 L = _____ mL (g) 1 g = _____ mg (h) 1 t = _____ kg

4. Choose the most sensible unit: mm, cm, m, km, mg, g, kg, t, mL or L.

(a) the length of your foot		(f) the mass of a bag of dog food	
(b) the distance of a marathon		(g) the water in a swimming pool	
(c) the thickness of a window pane		(h) the soy sauce in a recipe	
(d) the mass of a loaded gravel truck		(i) the height of a house	
(e) the medicine in one pill		(j) the mass of a pencil	

5. Circle the most reasonable measurement.

(a) the length of a canoe 5 mm 5 cm 5 m 5 km

(b) the mass of a newborn baby 3.5 mg 3.5 g 3.5 kg 3.5 t

(c) a can of pop holds 355 mL 35.5 L 355 L 3.55 mL

(d) a warm summer day in Calgary 2 °C 25 °C 60 °C 100 °C

(e) the height of a classroom ceiling 30 cm 3 m 30 m 3 km

6. Use a referent to estimate.

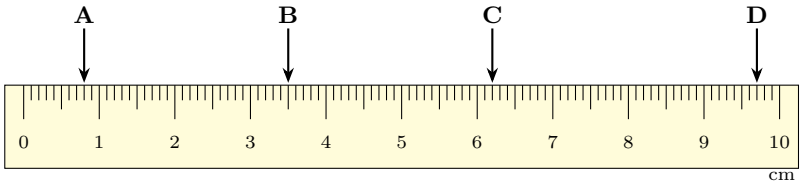
(a) Kenji's long step is about 1 m. He crosses a parking lot in 42 steps. About how wide is the lot?

(b) A door is about 2 m tall. A flagpole is about 5 doors tall. About how tall is the flagpole?

(c) A teaspoon holds about 5 mL. A recipe needs 3 teaspoons of vanilla. About how many millilitres is that?

(d) Name an object you could use as a referent for 1 kg, and one for 1 L.

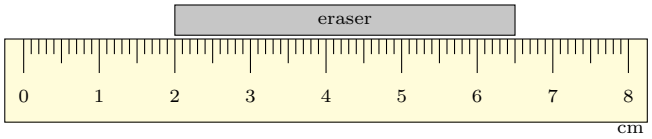
7. Read the ruler at A, B, C and D. Write each reading in centimetres and in millimetres.



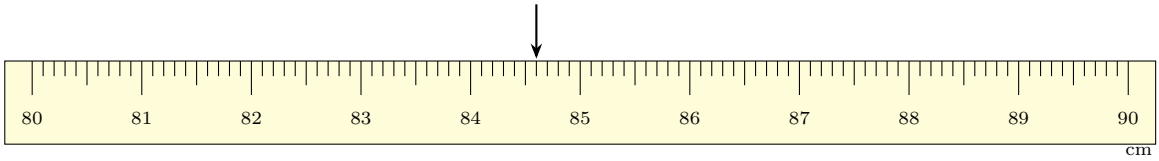
	cm	**mm**
A		
B		
C		
D		

8.

(a) How long is the eraser? Give your answer in cm and in mm.



(b) Liam measures a shelf with a tape measure. The end of the shelf is at the arrow. Write the length of the shelf in cm, in mm and in m.



9. Change each measurement.

(a) 5.2 m = _____ cm (b) 38 mm = _____ cm (c) 4.5 km = _____ m

(d) 180 cm = _____ m (e) 0.9 m = _____ mm (f) 2300 m = _____ km

(g) 3.2 kg = _____ g (h) 450 g = _____ kg (i) 1.75 L = _____ mL

(j) 500 mL = _____ L (k) 2 g = _____ mg (l) 6 mm = _____ m

10. Jun changed 4.6 m to centimetres. He wrote: 4.6 m = 0.046 cm.

(a) Without calculating, how can you tell his answer is wrong?

(b) What did Jun do wrong? Find the correct answer.

11. Write $<$, $>$ or $=$ in each circle. Change to the same unit first.

(a) 0.6 m ◯ 58 cm (b) 1.5 km ◯ 1500 m (c) 72 mm ◯ 7.5 cm

(d) 2.4 kg ◯ 2040 g (e) 0.3 L ◯ 350 mL (f) 90 cm ◯ 900 mm

12. The heights of four basketball players are Ava 1.62 m, Ben 158 cm, Cole 1655 mm and Dana 160.5 cm. Put them in order from shortest to tallest.

13. A 1.89 L jug of juice is poured into glasses that each hold 250 mL.

(a) Change 1.89 L to millilitres.

(b) How many glasses can be filled to the top?

(c) How much juice is left over?

14. A spool holds 5 m of ribbon. Each gift bow uses 35 cm of ribbon. How many bows can be made? How much ribbon is left?

15. A vet says Rex the dog needs 750 mg of a medicine each day. Each tablet has 0.25 g of the medicine.
(a) How many milligrams are in one tablet?

(b) How many tablets does Rex need each day?

16. 3.45 km is the same as
A. 345 m **B.** 3450 m **C.** 34 500 m **D.** 0.00345 m

17. The most reasonable mass for a loaf of bread is
A. 675 mg **B.** 675 g **C.** 675 kg **D.** 6.75 t

18. Numerical Response A hallway is 2.75 m wide. Its width in centimetres is _____ cm.

Lesson 3: The Imperial System and Reading a Tape Measure

Assignment

1. Fill in the blanks.

(a) 1 ft = _____ in (b) 1 yd = _____ ft (c) 1 yd = _____ in

(d) 1 mi = _____ ft (e) 1 mi = _____ yd

Use the facts above. Which is longer?

(f) 1 ft or 14 in? _____ (g) 1 yd or 40 in? _____ (h) 1 mi or 5000 ft? _____

2. Write each length with symbols or in words.

Write using the symbols $'$ and $''$.

(a) 4 feet 9 inches _____

(b) 11 inches _____

(c) 6 feet $3\frac{1}{2}$ inches _____

Write in words.

(d) $10'$ _____

(e) $5'\,6''$ _____

(f) $\frac{3}{4}''$ _____

3. Units and referents.

(a) Choose the best unit (inch, foot, yard or mile) for each.

the thickness of a textbook		the length of a pickup truck	
the distance driven in an hour on a US highway		the length of a living room	
the size of a TV screen		the length of a US football field	

(b) Circle the most reasonable length.

(i) the height of an adult $5\frac{1}{2}$ in $5\frac{1}{2}$ ft $5\frac{1}{2}$ yd $5\frac{1}{2}$ mi

(ii) the height of a basketball hoop 10 in 10 ft 10 yd 10 mi

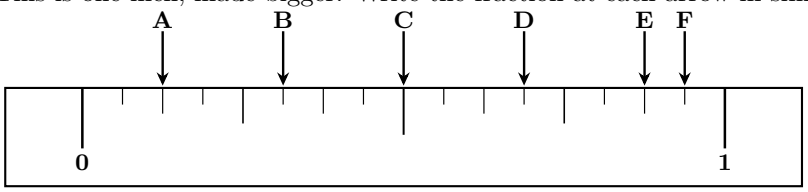
(iii) a 5 km run 3 in 3 ft 3 yd 3 mi

4. Write each in simplest form.

(a) $\frac{2}{16}$ _____ (b) $\frac{6}{16}$ _____ (c) $\frac{8}{16}$ _____ (d) $\frac{12}{16}$ _____ (e) $\frac{14}{16}$ _____ (f) $\frac{4}{8}$ _____

(g) Why can $\frac{9}{16}$ not be simplified?

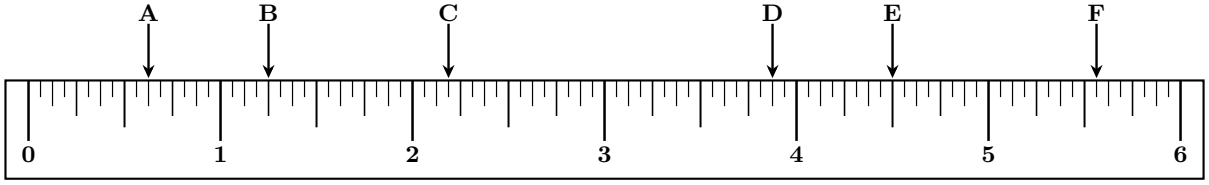
5. This is one inch, made bigger. Write the fraction at each arrow in simplest form.



A _____ B _____ C _____

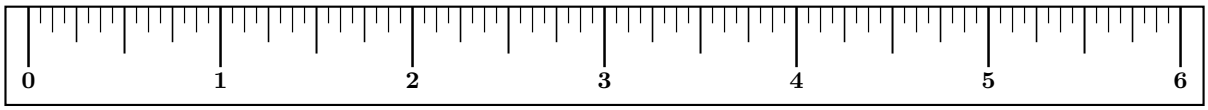
D _____ E _____ F _____

6. Read the length at each arrow. The ruler is real size.

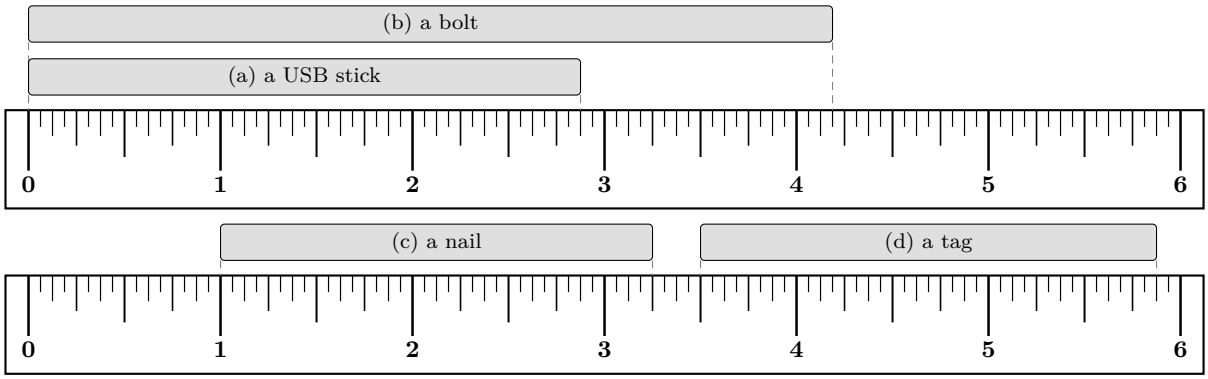


A _____ B _____ C _____ D _____ E _____ F _____

7. Draw an arrow on the ruler at each length. Label it with its letter.
P $1\frac{3}{8}$ in Q $2\frac{3}{4}$ in R $3\frac{1}{16}$ in S $4\frac{5}{16}$ in T $5\frac{1}{2}$ in

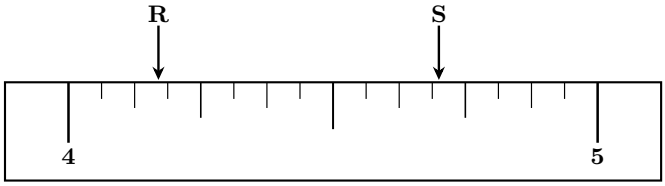


8. How long is each object? The rulers are real size.



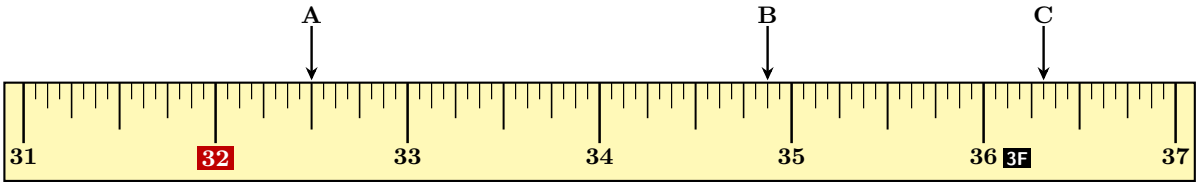
(a) _____ (b) _____ (c) _____ (d) _____

9. This is the part of a ruler from 4 in to 5 in, made bigger. Measure each arrow to the nearest $\frac{1}{16}$ in and to the nearest $\frac{1}{8}$ in.



	nearest $\frac{1}{16}$ in	nearest $\frac{1}{8}$ in
R		
S		

10. Read each arrow on this part of a tape measure (real size). Write it in inches, then in feet and inches.



A _____ = _____ B _____ = _____ C _____ = _____

11. A socket set has these sizes: $\frac{3}{8}''$, $\frac{5}{16}''$, $\frac{1}{2}''$, $\frac{7}{16}''$, $\frac{9}{16}''$, $\frac{1}{4}''$.
(a) Put them in order from smallest to largest.

(b) The $\frac{7}{16}''$ socket is too small for a nut. Which socket should Theo try next?

12. Four players on a basketball team are Amira $5' 8\frac{1}{2}''$, Ben $6' 1''$, Chloé $5' 11''$ and Dev $6' 0\frac{3}{4}''$.
(a) Order them from tallest to shortest.

(b) Who is closest to 6 ft tall? Explain.

13. Jayden is reading a ruler marked in sixteenths. The arrow is on the 6th line past the 2. He says the length is $2\frac{6}{8}$ in. What did he do wrong? What is the right length?

14. Rosa is cutting baseboard. She hooks her tape on the end of a long strip and makes two pencil marks, at $15\frac{1}{4}$ in and at $22\frac{5}{8}$ in. How long is the piece between the marks? Check your answer by adding.

15. Canada uses SI units. Give two places where Canadians still use imperial lengths. Why should someone who wants to work in a trade know how to read an imperial tape measure?

16. Which length is the longest?
A. $3\frac{5}{8}$ in **B.** $3\frac{3}{4}$ in **C.** $3\frac{11}{16}$ in **D.** $3\frac{7}{16}$ in

17. A tape measure shows $41\frac{1}{4}$ in. The 3F tag is at 36 in. In feet and inches, this length is
A. $4' 1\frac{1}{4}''$ **B.** $3' 5\frac{1}{4}''$ **C.** $3' 11\frac{1}{4}''$ **D.** $4' 5\frac{1}{4}''$

18. Numerical Response How many sixteenths of an inch are in $1\frac{3}{4}$ in?

19. Numerical Response The red numbers on a tape measure mark wall studs every 16 in: 16, 32, ... What is the 5th red number?

Lesson 4: Converting Length Within SI and Within Imperial

Assignment

1. Convert each length.

(a) 3.6 m to cm

(b) 72 mm to cm

(c) 5.08 km to m

(d) 940 cm to m

(e) 1.25 m to mm

(f) 8500 m to km

(g) 6 mm to m

(h) 0.35 km to cm

2. Write $<$, $>$ or $=$ in each box. Show how you know.

(a) 45 cm _____ 0.5 m

(b) 2300 m _____ 2.3 km

(c) 80 mm _____ 7 cm

(d) 0.09 km _____ 900 cm

3. Kayla wrote: 2.5 m = 0.025 cm. What mistake did she make? What is the correct answer?

4. A florist has a 25 m roll of ribbon. She uses 40 cm of ribbon on each bouquet.

(a) How many bouquets can she make with the roll?

(b) How much ribbon is left over?

5. A hiking trail in Kananaskis has three parts: 1.6 km, 850 m and 2.35 km.

(a) How long is the whole trail, in km and in m?

(b) Hailey has hiked 3.1 km so far. How many metres are left?

6. Convert each length.

(a) 7 ft to in

(b) 54 in to ft

(c) 24 ft to yd

(d) 2.5 yd to in

(e) 3 mi to ft

(f) 3520 yd to mi

7. The walking path around a park is 1760 ft long.

(a) Sam walks 3 laps. How far is that in feet, in yards and in miles?

(b) How many laps must Sam walk to go 2 mi?

8. Convert each length.

(a) $5'\,4''$ to inches

(b) $11'\,9''$ to inches

(c) $4'\,2\frac{1}{2}''$ to inches

(d) 39 in to ft and in

(e) 87 in to ft and in

(f) 200 in to ft and in

9. Convert. Round decimal feet to the nearest hundredth and inches to the nearest inch.

(a) $9'\,6''$ to decimal feet

(b) $4'\,9''$ to decimal feet

(c) $2'\,4''$ to decimal feet

(d) 12.25 ft to ft and in

(e) 6.5 ft to ft and in

(f) 2.8 ft to ft and in

(g) Leo says that 3.4 ft is the same as $3'\,4''$. Is he right? Explain, and give the correct length to the nearest inch.

10. Four friends compare their heights: Ava is $5'\,6''$, Ben is 67 in, Cole is 5.4 ft, and Dina is $5'\,5\frac{1}{2}''$. List them from shortest to tallest.

11. Find each answer in feet and inches, with fewer than 12 inches.

(a) $3'\,8'' + 6'\,7''$ (b) $12'\,2'' - 4'\,9''$ (c) $4 \times 1'\,10''$

(d) $5'\,6\frac{3}{4}'' + 3'\,7\frac{1}{2}''$ (e) $8' - 2'\,3\frac{1}{2}''$

12. Marco is putting three kitchen cabinets side by side along a wall that is $7'\,10''$ long. The cabinets are $2'\,6''$, $1'\,9''$ and $3'\,0''$ wide. Will they fit? If so, how much space is left?

13. Dana is building a bench from deck boards. The boards are 12 ft long and cost \$14.97 each. She needs 20 pieces that are each 40 in long.

(a) How many 40 in pieces can she cut from one board? How much is left over, in feet?

(b) How many boards must she buy? What will they cost?

14. Grace needs 3 pieces of ribbon, each $2'\,3''$ long. The store sells ribbon only in whole yards.

(a) How much ribbon does she need, in inches and in yards?

(b) How many yards must she buy? How much will be left over? Is it enough for one more piece?

15. How many metres are in 4.5 km?

A. 45 m **B.** 450 m **C.** 4500 m **D.** 45 000 m

16. Which length is the same as 76 in?

A. $6' 4''$ **B.** 6.4 ft **C.** $7' 6''$ **D.** 912 ft

17. $7' 9'' + 2' 6''$ equals

A. $10' 3''$ **B.** $9' 3''$ **C.** 9.15 ft **D.** 10.3 ft

18. Numerical Response A shelf is $5' 3''$ long. Its length in decimal feet, to the nearest hundredth, is _____ ft.

Lesson 5: Converting Length Between SI and Imperial

Assignment

1. Which length is longer? Explain how you know.

Pair	Longer	How I know
(a) 1 ft or 1 m		
(b) 3 in or 8 cm		
(c) 2 mi or 3 km		
(d) 100 yd or 100 m		
(e) 30 cm or 1 ft		

2. Estimate in your head. Show your thinking.

(a) A 20 cm pencil case, in inches

(b) A 6 ft couch, in centimetres

(c) 50 mi, in kilometres

(d) 200 km, in miles

3. Convert. Show the unit factor. Round to the nearest tenth.

(a) 14 in to cm

(b) 75 cm to in

(c) 9 ft to cm

(d) 400 m to yd

(e) 3.5 mi to km

(f) 42.2 km (a marathon) to mi

4. Kenji changed 18 in to centimetres. His work was $18 \div 2.54 \approx 7.1$ cm.

(a) Without a calculator, how can you tell his answer is wrong?

(b) What should he have done? Find the correct answer.

5. Ava says that 250 m is about 820 ft. Check her by converting 820 ft back to metres (use 1 ft = 30.48 cm). Is she right?

6. Change each fraction to a decimal first, then convert.

(a) A $\frac{3}{8}$ in wrench, in millimetres (nearest tenth)

(b) A board $5\frac{1}{2}$ in wide, in centimetres (nearest tenth)

(c) A phone $6\frac{1}{4}$ in tall, in centimetres (nearest tenth)

7. Change each metric length to inches, to the nearest $\frac{1}{16}$ in. Simplify the fraction.

(a) 6.5 cm

(b) 12 cm

(c) 30 cm

8. Complete the table. Round to the nearest centimetre or the nearest inch.

Height	Work	Converted
(a) $5'\,4''$		
(b) $6'\,1''$		
(c) 152 cm		
(d) 188 cm		

9. On a basketball team roster, Jalen is listed at $6'\,9''$ and Tomas at 198 cm. Who is taller, and by how many centimetres (nearest tenth)?

10. A drive-through has a sign: "Clearance $9'\,0''$". A delivery truck is 2.6 m tall. Will it fit? How much room is there, in centimetres?

11. Round to the nearest whole number.

(a) The speed limit on a highway in Idaho is 65 mph. What is that in km/h?

(b) A Canadian driver is going 100 km/h on that highway. Is she over or under the limit? By how much?

(c) A visitor from Texas sees a 50 km/h sign in Calgary. What is that in mph?

12. Ella's running app shows that she ran 4.2 mi in a fun run. Her three sponsors each pledged \$2.50 per kilometre.

(a) How far did Ella run, in kilometres, to the nearest tenth?

(b) Using your answer to (a), how much money did she raise?

13. A hardware store sells rope for \$1.29 per foot. A marine store sells the same rope for \$4.10 per metre. Which store has the lower price per metre? Show your work.

14. Nadia is sewing a costume for the school play. The pattern says she needs 3.2 m of fabric. The store sells the fabric for \$11.98 per yard, cut in steps of $\frac{1}{4}$ yd.

(a) How many yards must she buy?

(b) What is the cost, including 5% GST?

15. Some old school tracks were 440 yd around. Modern tracks are 400 m around.

(a) Which track is longer, and by how many metres (nearest tenth)?

(b) Four laps of the old track make 1 mile. A "metric mile" race is 1500 m. How many metres longer is a real mile? (Use 1 mi $\approx$ 1.609 km.)

16. A car is going 70 mph. About how fast is that in km/h?

A. 43 km/h **B.** 113 km/h **C.** 70 km/h **D.** 178 km/h

17. Which length is the **longest**?

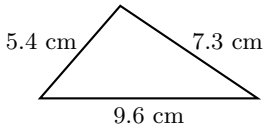
A. 2 m **B.** $6' 8''$ **C.** 2.2 yd **D.** 78 in

18. Numerical Response Kayla is 5 ft 7 in tall. Her height in centimetres, to the nearest centimetre, is _____ cm.

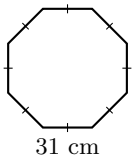
Lesson 6: Perimeter and Circumference

Assignment

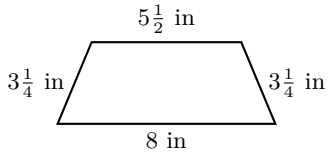
1. Find the perimeter of each shape.



(a) a triangle



(b) a stop sign (all sides equal)



(c) a wooden sign

2. Find the perimeter of each rectangle or square.

(a) a rectangle 14 m by 9 m

(b) a rectangle 6.8 cm by 4.25 cm

(c) a square with sides of $7\frac{3}{8}$ in

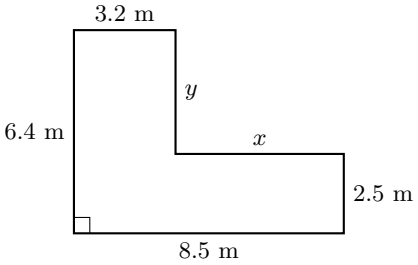
(d) a rectangle 5 ft 4 in by 3 ft 10 in (answer in feet and inches)

3. A rectangular patio is 12.5 m long and 6 m wide. Owen says its perimeter is $12.5 + 6 = 18.5$ m. Jada says it is $12.5 \times 6 = 75$ m.

(a) What mistake did each person make?

(b) Find the correct perimeter. Explain how a quick sketch or estimate shows that 18.5 m and 75 m are both wrong.

4.



All corners are right angles.

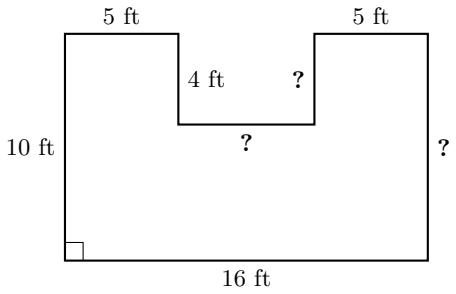
The diagram shows an L-shaped classroom.

(a) Find the missing side lengths x and y.

(b) Find the perimeter of the room.

(c) Check your answer with a second method.

5.



All corners are right angles.

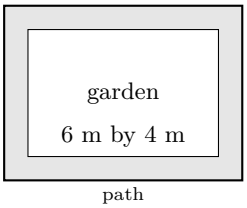
The diagram shows the outline of a U-shaped deck.

(a) Find the three missing side lengths (marked ?).

(b) Find the perimeter of the deck.

(c) Mia used the shortcut $2(16 + 10)$. Why is her answer wrong for this shape?

6.



A rectangular community garden plot is 6 m by 4 m. A gravel path 0.75 m wide goes all the way around it. Metal edging will go along the **outside** edge of the path.

(a) Find the length and width of the outside edge of the path.

(b) How much edging is needed?

7. A farmer puts an electric fence around a rectangular horse paddock 40 m by 32.5 m. The fence has 2 strands of wire, one above the other, all the way around.

(a) How much wire does the farmer need?

(b) The wire is sold in 250 m spools for $64.99. How many spools must the farmer buy, and what is the cost?

8. Find the circumference of each circle. Estimate first. Round to the nearest tenth.

(a) a dinner plate, $d = 12$ cm

(b) a hula hoop, $r = 45$ cm

(c) a round table top, $d = 2\frac{1}{2}$ ft

(d) a fire pit, $r = 0.63$ m

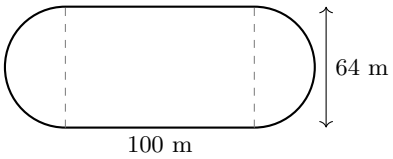
9. A car tire has a diameter of 64 cm.

(a) How far does the car roll when the tire turns once? Round to the nearest tenth of a centimetre.

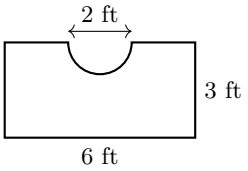
(b) How far does the car roll in 1000 turns of the tire? Give your answer in kilometres, to the nearest hundredth.

10. A jewellery maker bends silver wire into circles to make earrings. Each earring has 3 rings, and each ring has a diameter of 2.5 cm. How much wire does she need for one **pair** of earrings? Round to the nearest tenth of a centimetre.

11. Find the perimeter of each shape. Round to the nearest tenth.



(a) A running track has two straight sides 100 m long and a semicircle at each end. How far is one lap of the inside edge?



(b) A card table is a 6 ft by 3 ft rectangle with a half circle, 2 ft across, cut out of the middle of one long side for the dealer.

12. Kiran is fencing a rectangular vegetable garden 7.5 m by 4.8 m. There is no gate. Fencing costs $12.60 per metre.

(a) How much fencing does Kiran need?

(b) What is the cost of the fencing before tax?

(c) What is the cost with 5% GST?

13. A carpenter puts wood trim around the outside of a rectangular window, 4 ft 6 in by 3 ft 9 in. The trim is sold in 7 ft pieces for $9.47 each.

(a) Find the perimeter of the window in feet and inches.

(b) How many pieces must the carpenter buy, and what is the cost?

(c) How much trim will be left over?

14. Lucas is putting plastic edging around a circular flower bed with a diameter of 2.8 m. The edging is sold in 6 m rolls for $16.99 each.

(a) How much edging does he need? Round to the nearest tenth of a metre.

(b) How many rolls must he buy, and what do they cost?

15. The circumference of a circle with a radius of 5 cm, to the nearest tenth, is

A. 15.7 cm **B.** 31.4 cm **C.** 78.5 cm **D.** 62.8 cm

16. The perimeter of a rectangle 7.5 m long and 4 m wide is

A. 11.5 m **B.** 30 m **C.** 19 m **D.** 23 m

17. Numerical Response A circular trampoline has a diameter of 4.3 m. A padded cover goes all the way around its edge. The length of the cover, to the nearest tenth of a metre, is _____ m.

18. Numerical Response A rectangular poster is 2 ft 3 in by 1 ft 9 in. The perimeter of the poster, in inches, is _____ in.

Lesson 7: Rearranging Formulas: Finding a Missing Side

Assignment

1. What undoes each operation? Fill in the blank.

(a) add 7 $\rightarrow$ _____

(b) multiply by π $\rightarrow$ _____

(c) divide by 2 $\rightarrow$ _____

(d) add $2l$ $\rightarrow$ _____

2. Substitute, then undo to find the missing value. Check your answer.

(a) $P = 4s$; $P = 62$ cm. Find s.

(b) $P = a + b + c$; $P = 27$ in, $a = 8$ in, $b = 10$ in. Find c.

(c) $A = lw$; $A = 48$ m^2, $l = 8$ m. Find w.

(d) $C = \pi d$; $C = 31.4$ cm. Find d to the nearest tenth.

3. Rearrange each formula to find the letter shown.

(a) $C = \pi d$, so $d =$ _____

(b) $A = lw$, so $l =$ _____

(c) $P = 4s$, so $s =$ _____

(d) $C = 2\pi r$, so $r =$ _____

(e) $d = 2r$, so $r =$ _____

4. A hula hoop has a circumference of 251 cm. Estimate its diameter. Then find the diameter to the nearest centimetre.

5. A round steel fire pit ring has a circumference of 13 ft.

(a) Find its radius, to the nearest tenth of a foot.

(b) Find its diameter, to the nearest tenth of a foot.

6. (a) A rectangular sheet of drywall covers 32 ft^2. It is 8 ft long. How wide is it?

(b) A rectangular parking stall has an area of 14.4 m^2. It is 2.5 m wide. How long is it?

7. Fill in the blanks to rearrange $P = 2l + 2w$ to find w.

Step 1: Subtract _____ from both sides: $P - 2l =$ _____

Step 2: Divide both sides by _____: $w =$ _____

8. Hiro has 46 m of fencing for a rectangular garden. The garden will be 9 m wide. How long will it be? Solve by substituting first, then check.

9. A landscaper puts edging around rectangular garden plots. Use $w = (P - 2l) \div 2$ to complete the table.

	Edging (P)	**Length (l)**	**Work**	**Width (w)**
(a)	40 m	12 m		
(b)	25 m	7.5 m		
(c)	100 cm	30 cm		
(d)	18 ft	$5\frac{1}{4}$ ft		

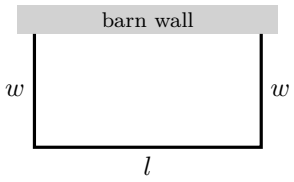
10. Rosa is sewing binding (a fabric strip) all the way around the edge of a rectangular quilt. She has exactly 270 in of binding. The quilt is $72\frac{1}{2}$ in long. How wide can it be?

11. A rectangular dance floor has a perimeter of 60 ft and is 18 ft long. Here is Ella's work:

$60 = 2(18) + 2w$, so $60 - 18 = 2w$, $42 = 2w$, $w = 21$ ft.

Check her answer by substituting back. Explain her mistake and find the correct width.

12. Owen is building a rectangular goat pen against the side of a barn. The barn wall is one length, so the fence goes on only three sides: one length and two widths.



(a) Fence costs \$15.50 per metre. Owen spends \$620.00. How many metres of fence does he buy?

(b) The fence length is $F = l + 2w$. Rearrange it to find w.

(c) The pen will be 18 m long. How wide will it be? Check your answer.

13. Ana has a 9.5 m roll of flexible edging for a flower bed. She wants to use all of it.

(a) If the bed is a circle, what is its diameter, to the nearest tenth of a metre?

(b) If the bed is a square, how long is each side, to the nearest hundredth of a metre?

14. Which is $P = 2l + 2w$ rearranged to find w?

A. $w = P - 2l$ **B.** $w = (P - 2l) \div 2$ **C.** $w = P \div 2 - 2l$ **D.** $w = (P + 2l) \div 2$

15. A round swimming pool has a circumference of 18.85 m. Its diameter is about

A. 59.2 m **B.** 3.0 m **C.** 6.0 m **D.** 9.4 m

16. Numerical Response A rectangular sign has a perimeter of 37 ft. It is 10.75 ft long. Its width, in feet, is _____.

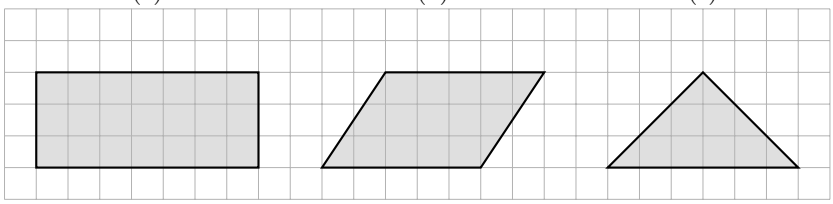
17. Numerical Response A round clock face has a circumference of 81.7 cm. Its radius, to the nearest tenth of a centimetre, is _____ cm.

Lesson 8: Area of 2-D Shapes

Assignment

1. Each small square stands for 1 cm^2. Find the area of each shape by counting. For (b) and (c), check with a formula.

(a) (b) (c)



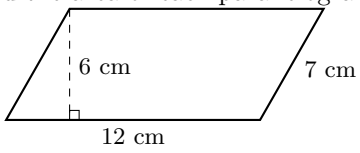
(a) _____ (b) _____ (c) _____

2. Find the area of each rectangle or square.

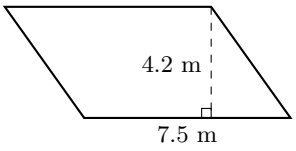
(a) 8 m by 5 m (b) 6.4 cm by 2.5 cm (c) a square, sides 9 in (d) 6 ft by $4\frac{1}{2}$ ft

3. Find the area of each parallelogram.

(a)



(b)



4. Find the area of each triangle.

(a) base 10 cm, height 7 cm (b) base 4.6 m, height 3 m (c) base 9 in, height $5\frac{1}{2}$ in

5. Find the area of each trapezoid.

(a) parallel sides 5 cm and 9 cm, height 4 cm

(b) parallel sides $2\frac{1}{2}$ ft and 4 ft, height 3 ft

6. Complete the table. Round areas to the nearest tenth.

	Radius	**Diameter**	**Area**
(a)	3 cm		
(b)		10 m	
(c)	2.5 ft		
(d)		$1\frac{1}{2}$ in	

7. A painter is painting a wall that is 4.8 m long and 2.6 m high.

(a) Find the area of the wall.

(b) One litre of paint covers about 10 m^2. Is 1 L enough for one coat? Explain.

8. A sailmaker in Kelowna is making a triangular sail with a base of 2.8 m and a height of 4.5 m. The sail cloth costs \$18.75 per m^2.

(a) Find the area of the sail.

(b) Find the cost of the cloth, to the nearest cent.

9. A round coffee table has a diameter of 42 in. A glass shop charges \$0.35 per square inch to cut a glass top for it.

(a) Find the area of the top, to the nearest square inch.

(b) Use your rounded area to find the cost of the glass.

10. Complete the table. Round to the nearest tenth if needed.

Change	**Work**	**Answer**
(a) 4 ft^2 to in^2		
(b) 1008 in^2 to ft^2		
(c) 6 yd^2 to ft^2		
(d) 2.4 m^2 to cm^2		
(e) 45 000 cm^2 to m^2		
(f) 20 m^2 to ft^2		

11. A landscaper is laying sod on a backyard in Cochrane that is 12.4 m long and 8.5 m wide. Sod costs \$3.85 per m^2.

(a) Find the area of the yard.

(b) Find the cost of the sod with 5% GST.

12. Carpet is often priced by the square yard. A living room is 18 ft long and 13 ft 6 in wide. The carpet costs \$28.95 per yd^2.

(a) Find the area in ft^2, then in yd^2.

(b) Find the cost of the carpet before tax.

13. Diego is tiling a kitchen floor that is 3.6 m by 2.7 m. Each tile is 30 cm by 30 cm. Tiles come in boxes of 10 for \$32.49 a box.

(a) Find the area of the floor in m^2.

(b) Find the area of one tile in m^2. (Change 30 cm to metres first.)

(c) How many tiles cover the floor? Add 10% for broken and cut tiles.

(d) How many boxes must Diego buy, and what do they cost before tax?

14. A rectangular rug has an area of 6 m^2. It is 2.5 m long. How wide is it? Check your answer.

15. A triangle has a base of 10 m and a height of 6 m. Its area is

A. 16 m^2 **B.** 30 m^2 **C.** 60 m^2 **D.** 120 m^2

16. How many square inches are in 2 ft^2?

A. 24 in^2 **B.** 144 in^2 **C.** 288 in^2 **D.** 576 in^2

17. Numerical Response A round hot tub cover has a radius of 4.2 ft. Its area, to the nearest tenth of a square foot, is _____ ft^2.

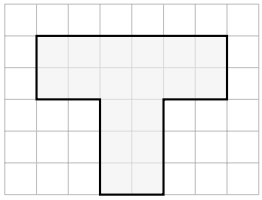
18. Numerical Response A bedroom is 21 ft long and 15 ft wide. The number of square yards of carpet needed to cover the floor is _____ yd^2.

Lesson 9: Composite and Irregular Shapes

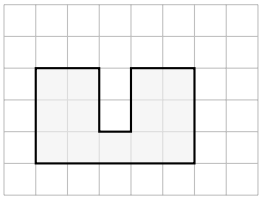
Assignment

1. Each square on the grid is 1 cm by 1 cm. Find the area of each shape by counting squares. Then check by splitting the shape into rectangles.

(a)

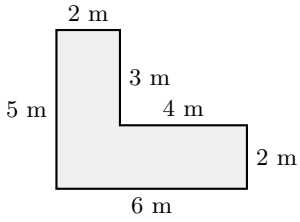


(b)

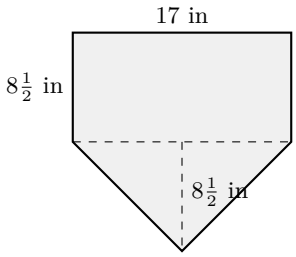


2. Find the area of each composite shape. All the corners that look square are square.

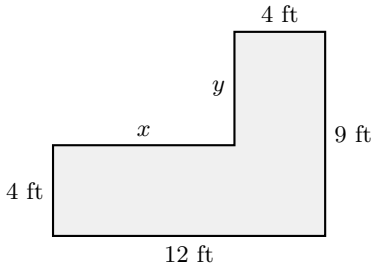
(a)



(b) A baseball home plate



3. The plan shows an L-shaped deck. All the corners are square.

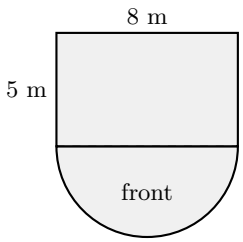


(a) Find x and y.

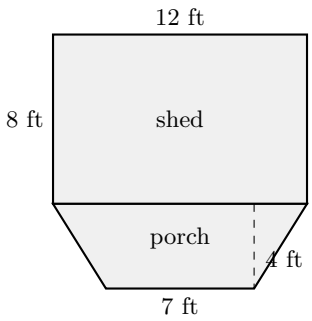
(b) Find the area of the deck by splitting and adding.

(c) Check your answer using whole minus cut-out.

4. The floor of a school stage is a rectangle 8 m wide and 5 m deep, with a semicircle across the front. Find the area of the stage floor, to the nearest tenth of a square metre.



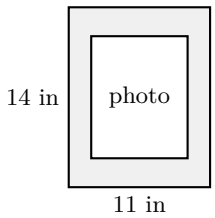
5. Diego is building the floor of a shed with a porch. The shed is a rectangle 12 ft by 8 ft. The porch is a trapezoid.



(a) Find the total floor area. (Trapezoid: $A = \frac{1}{2}(a + b)h$)

(b) A sheet of plywood is 4 ft by 8 ft. How many sheets does Diego need? (Ignore waste.)

6. A picture mat is a rectangle 11 in by 14 in. A rectangle $7\frac{1}{2}$ in by $9\frac{1}{2}$ in is cut out of the middle for the photo. Find the area of the mat that is left.



7. A square patio is 6 m by 6 m. A round fire pit 1.2 m across sits in the middle. The rest of the patio is covered with paving stones.

(a) Find the area covered with paving stones, to the nearest hundredth of a square metre.

(b) One pallet of paving stones covers 10 m^2. How many pallets are needed?

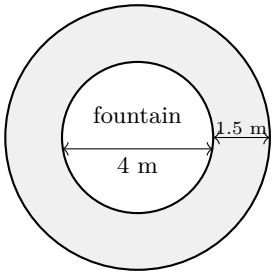
8. Aisha is painting a feature wall 14 ft long and 9 ft high. The wall has two windows, each 3 ft wide and 4 ft 6 in high.

(a) Make a sketch of the wall and label it.

(b) Find the area she will paint.

(c) She will paint two coats. One can covers about 90 ft^2. How many cans does she need?

9. A round fountain is 4 m across. A path 1.5 m wide goes all the way around it.

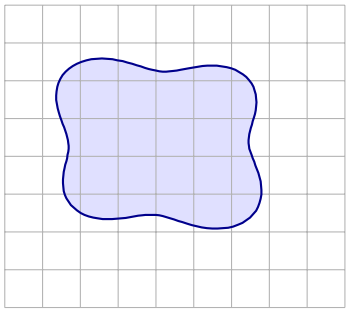


(a) What is the radius of the fountain? What is the radius of the outside edge of the path?

(b) Find the area of the path, to the nearest hundredth of a square metre.

(c) Stamped concrete costs $42.00 per m^2. Using your rounded area, find the cost of the path.

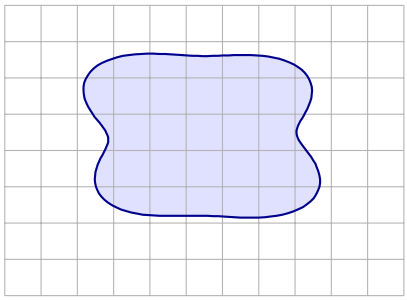
10. The grid shows a flower garden in a park. Each square is 1 m by 1 m.



Full squares	
Part squares	
Part squares ÷ 2	
Estimate of area	

Estimate the area of the garden by completing the table.

11. A golf course map shows a pond. Each square stands for 2 m by 2 m.

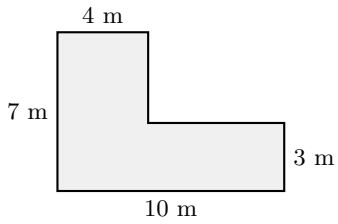


(a) What area does one square stand for?

(b) Estimate the area of the pond.

(c) Draw the smallest rectangle (on grid lines) around the pond. Use it to check that your estimate is reasonable.

12. Kayla found the area of this L-shaped room. Here is her work:



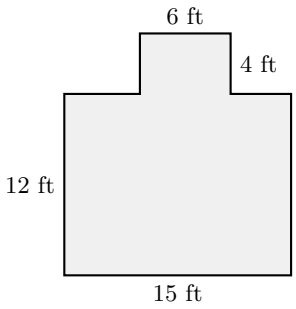
Bottom part: $10 \times 3 = 30\ m^2$
Left part: $4 \times 7 = 28\ m^2$
Total: $30 + 28 = 58\ m^2$

(a) What mistake did Kayla make?

(b) Find the correct area.

(c) Check your answer using whole minus cut-out.

13. A bedroom is a rectangle 15 ft by 12 ft with a reading nook 6 ft by 4 ft on one side.

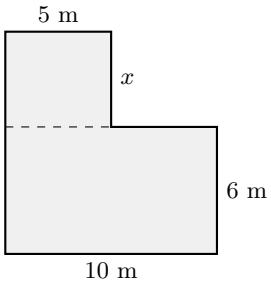


(a) Find the area of the floor in square feet.

(b) Carpet is sold by the square yard ($1\ \text{yd}^2 = 9\ \text{ft}^2$). Change the area to square yards, to the nearest hundredth.

(c) The store sells only whole square yards, at \$34.00 per yd^2. What will the carpet cost (before tax)?

14. A garden is made of a rectangle 10 m by 6 m and a rectangle 5 m wide, joined as shown. The total area of the garden is 82.5 m^2. Find the length x.



15. The area of a semicircle with a diameter of 10 cm, to the nearest hundredth, is

A. 78.54 cm^2 **B.** 39.27 cm^2 **C.** 157.08 cm^2 **D.** 15.71 cm^2

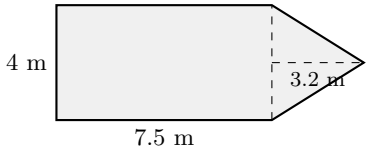
16. A lawn is 20 m by 15 m. A square pool 5 m by 5 m is in the middle. Which calculation gives the area of the lawn?

A. $20 \times 15 + 5 \times 5$ **B.** $(20 - 5) \times (15 - 5)$ **C.** $20 \times 15 - 5 \times 5$ **D.** $2(20) + 2(15) - 4(5)$

17. On a grid, a puddle covers 12 full squares and 10 part squares. Each square stands for 4 m^2. The best estimate of the puddle's area is

A. 48 m^2 **B.** 88 m^2 **C.** 17 m^2 **D.** 68 m^2

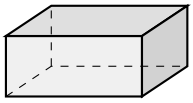
18. Numerical Response A garden plot is a rectangle 7.5 m by 4 m with a triangle on one 4 m end. The triangle has a base of 4 m and a height of 3.2 m. The area of the garden plot, to the nearest tenth of a square metre, is _____ m^2.



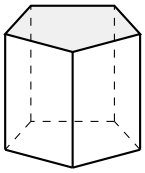
Lesson 10: Nets and Surface Area of Prisms

Assignment

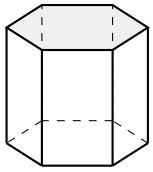
1. Complete the table for each prism.



(a)



(b) a 5-sided base

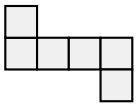


(c) a 6-sided base

	(a)	**(b)**	**(c)**
Name of the prism			
Number of faces			
Number of edges			
Number of vertices			

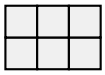
2. Which patterns are nets of a cube? Write yes or no.

(a)



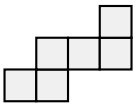
_____

(b)



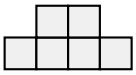
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(c)



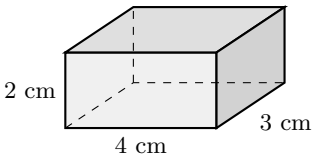
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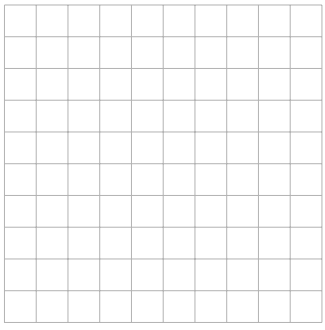
(d)



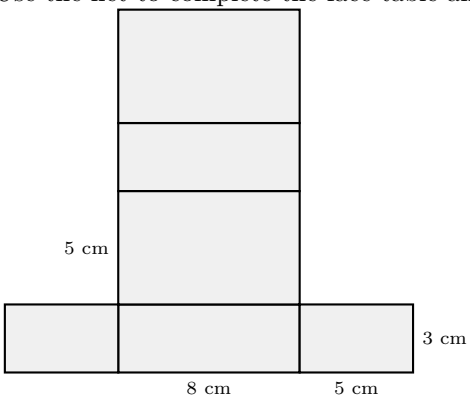
_____

3. A small box is 4 cm long, 3 cm wide and 2 cm high. Draw a net of the box on the grid (each square is 1 cm long). Then find the surface area by counting squares.





4. Use the net to complete the face table and find the surface area of the box.



Faces	**Size**	**Area of one**	**Area of two**
top & bottom			
front & back			
two ends			
		Surface area	

5. Find the surface area of each cube or rectangular prism.

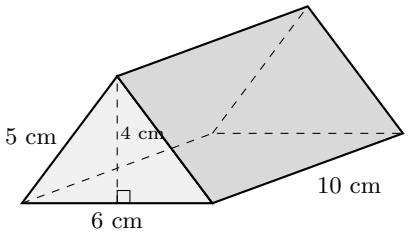
(a) a cube with edges 4 cm

(b) a cube with edges $3\frac{1}{2}$ in

(c) a prism 2.5 m long, 1.2 m wide, 0.8 m high

(d) a prism 2 ft long, 1 ft 6 in wide, 9 in high (answer in ft^2)

6. Find the surface area of the triangular prism. Its ends are triangles with a base of 6 cm and a height of 4 cm; both sloping sides are 5 cm.



7. A rubber curb ramp helps bikes and cars roll up onto a driveway. It is a triangular prism. Its ends are right triangles with sides 5 in, 12 in and 13 in, and the ramp is 24 in wide.

(a) Find the total surface area of the ramp.

(b) The bottom (12 in by 24 in) lies on the ground. The rest is painted yellow. Find the area that is painted.

8. Ella is wrapping a gift box that is 30 cm long, 20 cm wide and 10 cm high.

(a) Find the surface area of the box.

(b) Her sheet of wrapping paper is 70 cm by 50 cm. Does it have enough area?

(c) Why does wrapping a box always use more paper than its surface area?

9. A glass fish tank has no top. It is 24 in long, 12 in wide and 16 in high.

(a) Which faces are made of glass? Find the area of glass in in^2.

(b) Change your answer to ft^2.

10. A bakery shows its menu on a cardboard stand shaped like a triangular prism. The cardboard covers the two ends and the two sloping sides; the bottom is open. The ends are triangles with a base of 30 cm and a height of 20 cm, the sloping sides are 25 cm, and the stand is 80 cm long.

(a) Find the area of cardboard in cm^2.

(b) Change your answer to m^2.

11. Marco is putting drywall on the walls and ceiling of his garage workshop. The garage is 20 ft long, 12 ft wide and 9 ft high. He does not cover the big garage door (9 ft by 7 ft) or the side door (3 ft by 7 ft).

(a) Find the area of the four walls and the ceiling.

(b) Subtract the two doors.

(c) A sheet of drywall is 4 ft by 8 ft. How many sheets must he buy?

(d) Each sheet costs \$16.48. Find the total cost with 5% GST.

12. Jae was asked for the surface area of a box 5 cm long, 4 cm wide and 3 cm high. Jae wrote: $SA = 5 \times 4 \times 3 = 60$ cm^2. Explain the mistake, then find the correct surface area.

13. A cube has a surface area of 150 cm^2. How long is each edge? Check your answer.

14. What is the surface area of a cube with edges 3 m long?

A. 9 m^2 **B.** 27 m^2 **C.** 36 m^2 **D.** 54 m^2

15. An open box (no top) is 6 in long, 4 in wide and 2 in high. How much cardboard does it use?

A. 40 in^2 **B.** 48 in^2 **C.** 64 in^2 **D.** 88 in^2

16. The walls of a hall have an area of 95 m^2. They need 2 coats of paint. One can of paint covers 40 m^2. How many cans must be bought?

A. 5 cans **B.** 2 cans **C.** 3 cans **D.** 4 cans

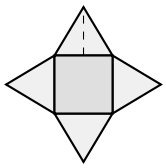
17. Numerical Response A wooden crate is 2.5 m long, 2 m wide and 1.2 m high. Its surface area, to the nearest tenth of a square metre, is _____ m^2.

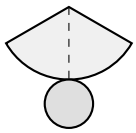
18. Numerical Response A moving box is 36 in long, 24 in wide and 18 in high. Its surface area, in square feet, is _____ ft^2.

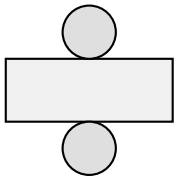
Lesson 11: Surface Area of Cylinders, Pyramids and Cones

Assignment

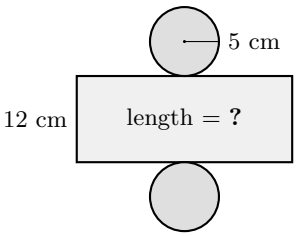
1. Name the solid that each net folds into.

(a) 

(b) 

(c) 

2. A cylinder has a radius of 5 cm and a height of 12 cm. Here is its net.



Round each answer to the nearest tenth.

(a) Find the length of the rectangle.

(b) Find the area of the rectangle.

(c) Find the area of one circle.

(d) Find the surface area of the cylinder.

3. Find the surface area of each closed cylinder, to the nearest tenth.

(a) radius 3 m, height 8 m

(b) diameter 10 in, height 4 in

(c) radius 1.5 ft, height $2\frac{1}{2}$ ft

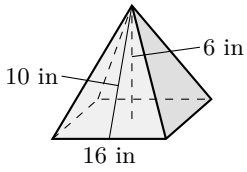
4. A juice can has a diameter of 6.6 cm and a height of 12 cm. The label covers the whole curved side.

(a) Find the area of the label, to the nearest tenth.

(b) The printer wants to cut each label from a sheet that is 20 cm long and 12 cm wide. Will the label fit? Explain.

5. Find the surface area of each square pyramid.

(a) base edge 6 cm, slant height 9 cm (b) base edge 2.5 m, slant height 4 m (c)



6. Find the surface area of each closed cone, to the nearest tenth.

(a) radius 4 cm, slant height 7 cm (b) diameter 12 ft, slant height 10 ft (c) radius 0.5 m, slant height 1.3 m

7. A paper party hat is a cone with no base. It is 16 cm across the bottom and has a slant height of 20 cm.

(a) How much paper is in one hat, to the nearest square centimetre?

(b) Leah is making 30 hats for a party. How much paper is that, in m^2, to the nearest tenth? ($1\ m^2 = 10\,000\ cm^2$)

8. A metal bucket has a bottom but no lid. It has a diameter of 30 cm and a height of 28 cm. How much metal is in the bucket, to the nearest square centimetre?

9. The roof of a square gazebo is a square pyramid. Each side of the base is 12 ft and the slant height is 9 ft. One bundle of shingles covers 33.3 ft^2 and costs $38.97.

(a) Find the area of the roof. (A roof has no base.)

(b) How many bundles are needed?

(c) What do the bundles cost with 5% GST?

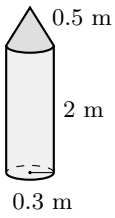
10. A jewellery store has a glass display case shaped like a square pyramid. It sits on a table, so it has no bottom. The base edge is 45 cm and the slant height is 50 cm.

(a) Find the area of glass in cm^2, then in m^2.

(b) The glass costs \$85.00 per m^2. Find the cost of the glass.

11. A company is choosing between two closed cans that hold the same amount of juice. Can A has a radius of 4 cm and a height of 10 cm. Can B has a radius of 5 cm and a height of 6.4 cm. Which can uses less metal? Show your work.

12. A giant crayon for a school display is a cylinder with a cone on top. The cylinder has a radius of 0.3 m and a height of 2 m. The cone has a slant height of 0.5 m. Every outside surface gets painted, including the flat bottom. Find the area to paint, to the nearest tenth.



13. Tyler found the surface area of a closed cylinder with a diameter of 8 cm and a height of 5 cm. Find his mistake, then find the correct answer to the nearest tenth.

$SA = 2 \times \pi \times 8^2 + 2 \times \pi \times 8 \times 5 = 402.12 + 251.33 = 653.45 \text{ cm}^2$

14. A cylinder has a radius of 3 cm and a height of 10 cm. The area of its curved side, to the nearest tenth, is

A. 94.2 cm^2 **B.** 188.5 cm^2 **C.** 245.0 cm^2 **D.** 282.7 cm^2

15. A closed cone has a radius of 5 cm, a height of 12 cm and a slant height of 13 cm. Its surface area, to the nearest tenth, is

A. 204.2 cm^2 **B.** 267.0 cm^2 **C.** 282.7 cm^2 **D.** 722.6 cm^2

16. Numerical Response A square pyramid has a base edge of 5 m and a slant height of 8 m. Its surface area is _____ m^2.

17. Numerical Response A closed cylinder has a radius of 2 ft and a height of 3 ft. Its surface area, to the nearest tenth of a square foot, is _____ ft^2.

Lesson 12: Unit 3 Review

Assignment

This review is set up like the unit test. Show your work on the written questions. Use the conversion numbers on the summary page and the $\boxed{\pi}$ key.

Part A: Written Response

1. Fill in the blanks.

(a) 4.6 m = _____ cm (b) 85 mm = _____ cm (c) 3200 m = _____ km

(d) 0.75 km = _____ m (e) 2.4 cm = _____ mm (f) 1250 mm = _____ m

2. Fill in the blanks.

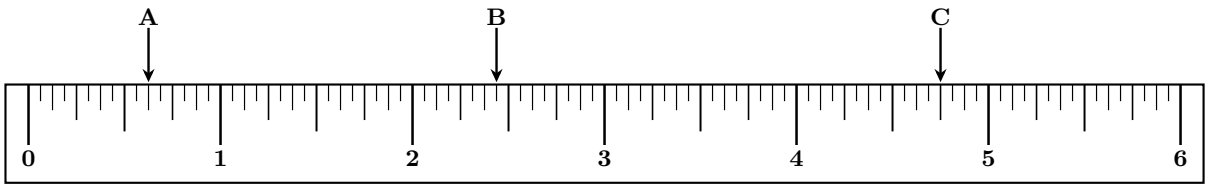
(a) 4 ft = _____ in (b) 15 ft = _____ yd (c) 2 mi = _____ ft

(d) $6' 3''$ = _____ in (e) 50 in = _____ ft _____ in (f) 2 yd = _____ in

3. Find each answer in simplest form.

(a) $1\frac{7}{8} + 3\frac{3}{4}$ (b) $6\frac{1}{4} - 3\frac{5}{8}$ (c) $3 \times 2\frac{5}{8}$

4. Read the length at each arrow. The ruler is real size.



A _____ B _____ C _____ (d) How far is it from B to C? _____

5. Convert. Round to the nearest tenth unless told otherwise.

(a) A board 18 in wide, in centimetres.

(b) A 250 cm roll of ribbon, in inches.

(c) A 42 km marathon, in miles.

(d) Ava's height of $5'\,4''$, in centimetres (to the nearest centimetre).

(e) A highway sign in Montana says the speed limit is 65 mph. Jas sets his cruise control to 100 km/h. Is he under the limit?

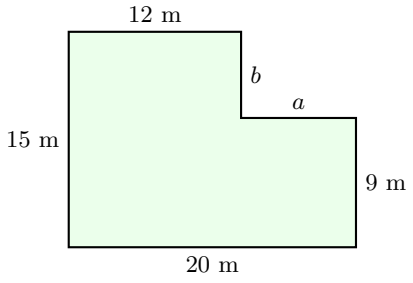
6. Theo has an 8 ft board. He cuts off two pieces, $3'\,7''$ and $2'\,9''$ long. (Ignore the saw cuts.)

(a) What is the total length of the two pieces?

(b) How much of the board is left?

(c) Can he cut a third piece $1'\,10''$ long from what is left? Explain.

7. The diagram shows a back yard. All the corners are square.



(a) Find the missing lengths a and b.

(b) Find the perimeter of the yard.

(c) Fencing costs \$18.75 per metre. Find the cost to fence the yard, leaving a 4 m gap for the driveway.

8. A round above-ground pool is 5.5 m across.

(a) Find the circumference of the pool, to the nearest tenth of a metre.

(b) A solar cover lies flat on the water. Find its area, to the nearest tenth.

(c) Which answer is in square units? Why?

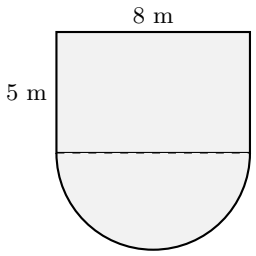
9. Rearrange the formula to find the missing value. Check each answer.

(a) A rectangle has a perimeter of 54 ft and a width of 9.5 ft. Find its length. ($P = 2l + 2w$)

(b) A round flower bed has a circumference of 12.5 m. Find its diameter, to the nearest hundredth. ($C = \pi d$)

(c) A rug has an area of 9.6 m^2 and a length of 3.2 m. Find its width. ($A = lw$)

10. A school stage is a rectangle 8 m wide and 5 m deep, with a half circle on the front. The half circle is 8 m across.



(a) Find the area of the stage, to the nearest hundredth.

(b) One can of floor paint covers 15 m^2. How many cans are needed for one coat?

11. Maya's bedroom floor is a rectangle $13' 6''$ by $11' 0''$.

(a) Find the area of the floor in square feet.

(b) Change the area to square yards.

(c) Carpet costs \$36.80 per square yard. What does carpet for the room cost before tax?

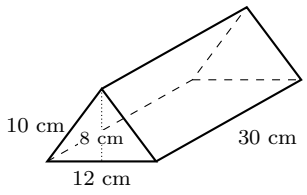
12. A wooden toy bin is a rectangular prism 1.2 m long, 0.5 m wide and 0.4 m tall. It has **no top**.

(a) Complete the face table and find the outside surface area.

Faces	**Size**	**Area**
bottom		
front and back		
2 ends		
	Surface area	

(b) A waterproof stain costs \$8.50 for each square metre covered. Find the cost to stain the outside.

13. A gift box is a triangular prism. Each triangle end has sides of 10 cm, 10 cm and 12 cm and a height of 8 cm. The box is 30 cm long.



(a) Sketch a net of the box. Label every face with its size.

(b) Find the surface area of the box.

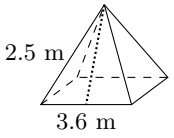
14. A farm water tank is a cylinder 2.4 m across and 3 m tall. The side and the top will be painted (not the bottom).

(a) Find the area to be painted, to the nearest hundredth.

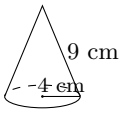
(b) One can of paint covers 10 m^2. How many cans are needed for one coat?

15.

(a) The roof of a bell tower is a square pyramid with a base 3.6 m on each side. Each triangle has a slant height of 2.5 m. Only the 4 triangles get shingles. Find the area to shingle.



(b) One bundle of shingles covers 3.1 m^2. How many bundles are needed?



(c) A paper water cup is a cone with a radius of 4 cm and a slant height of 9 cm. It has no lid. How much paper is in the cup, to the nearest square centimetre?

Part B: Multiple Choice

16. Which is the most reasonable width for a classroom door?

A. 90 cm **B.** 90 mm **C.** 9 m **D.** 0.9 km

17. $3'\,8'' + 2'\,7''$ is equal to

A. $6'\,3''$ **B.** $6'\,5''$ **C.** $5'\,3''$ **D.** $6'\,15''$

18. A rug covers 6 yd^2. How many square feet is that?

A. 54 ft^2 **B.** 18 ft^2 **C.** 72 ft^2 **D.** 2 ft^2

19. A round patio is 10 m across. Its area is about

A. 78.5 m^2 **B.** 31.4 m^2 **C.** 314.2 m^2 **D.** 15.7 m^2

20. A trapezoid has parallel sides of 6 cm and 10 cm and a height of 4 cm. Its area is

A. 32 cm^2 **B.** 64 cm^2 **C.** 240 cm^2 **D.** 20 cm^2

21. A spill on a grid covers 18 full squares and 10 part squares. Each square is 1 m^2. The best estimate of its area is

A. 23 m^2 **B.** 28 m^2 **C.** 18 m^2 **D.** 19 m^2

22. A cube has edges of 4 cm. Its surface area is
A. 96 cm^2 **B.** 64 cm^2 **C.** 16 cm^2 **D.** 80 cm^2

23. A cone has a radius of 3 cm and a slant height of 5 cm. Its total surface area (with the base) is about
A. 75.4 cm^2 **B.** 47.1 cm^2 **C.** 28.3 cm^2 **D.** 207.3 cm^2

Part C: Numerical Response

24. Numerical Response A doorway is 30 in wide. Its width in centimetres, to the nearest tenth, is _____ cm.

25. Numerical Response A bike wheel is 66 cm across. Its circumference, to the nearest centimetre, is _____ cm.

26. Numerical Response A rectangle has a perimeter of 38 m and a length of 11.5 m. Its width is _____ m.

27. Numerical Response A closed cylinder has a radius of 5 cm and a height of 12 cm. Its surface area, to the nearest square centimetre, is _____ cm^2.
